

Tadahiro Kuroda

professor

基本信息

姓名: Tadahiro Kuroda

所属院校: 东京大学

地区: Japan

年龄范围: 50岁以上

海外经历: 有

企业经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

集成电路设计与集成系统

微电子科学与工程

人工智能

电磁场与无线技术

电子与计算机工程

通信工程

研究领域

低功耗电路与能量管理

微电子器件与制造

深度学习与神经网络

个人简介

Tadahiro Kuroda is a distinguished professor at The University of Tokyo, specializing in integrated circuits and electronics. He received his PhD from the University of Tokyo in 1982 and has held significant positions at Toshiba Corporation and Keio University. His research focuses on advancing technology for future societies through integrated circuits, including near-field coupling,

energy-efficient computing, smart sensors, AI, and power electronics. Prof. Kuroda has authored 30 books, presented numerous papers at leading conferences, and holds over 200 patents. He is an IEEE Fellow, IEEE Distinguished Lecturer, and a member of the IEEE SSCS Administrative Committee. He currently chairs the VLSI Symposia and A-SSCC committees.

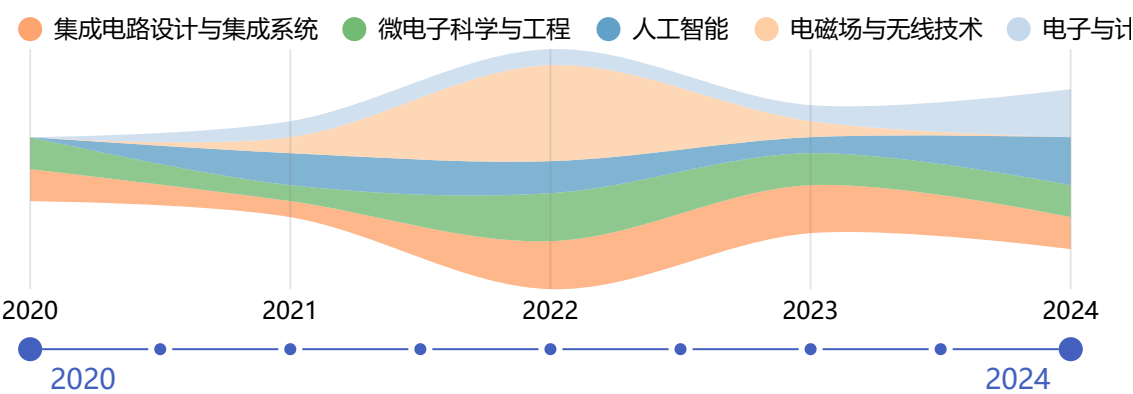
工作经历

The University of Tokyo - Professor -

教育经历

The University of Tokyo - Ph.D. -

近年发表的论文



A 28-nm 0.8M-weights/mm2 9.1-TOPS/mm2 SRAM-Based All-Analog Compute-In-Memory Using Fine-Grained Structured Pruning with Adaptive-Ranging ADC
2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024
Shiba, Kota; Zhan, Zhijie; Nii, Koji; Wang, Yih; Chang, Tsung-Yung Jonathan; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro
被引用次数: 0 2024

A 0.13 mJ/Prediction CIFAR-100 Fully Synthesizable Raster-Scan-Based Wired-Logic Processor in 16-nm FPGA
IEICE TRANSACTIONS ON ELECTRONICS
Li, Dongzhu; Zhan, Zhijie; Sumikawa, Rei; Hamada, Mototsugu; Kosuge, Atsutake; Kuroda, Tadahiro
被引用次数: 0 2024

Agile-X: A Structured-ASIC Created With a Mask-Less Lithography System Enabling Low-Cost and Agile Chip Fabrication
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kosuge, Atsutake; Sumi, Hirofumi; Shimamoto, Naonobu; Ochiai, Yukinori; Inoue, Yurie; Amano, Hideharu; Mogami, Tohru; Mita, Yoshio; Ikeda, Makoto; Kuroda, Tadahiro

被引用次数: 0

2024

Introduction of thick AlN coating on Si for 3D-IC thermal management

JAPANESE JOURNAL OF APPLIED PHYSICS

Ninomiya, Takeki; Takagi, Takeshi; Mori, Masakazu; Niwa, Masaaki; Kuroda, Tadahiro

被引用次数: 0

2024

A183.4-nJ/Inference 152.8-μW 35-Voice Commands Recognition Wired-Logic Processor Using Algorithm-Circuit Co-Optimization Technique

IEEE SOLID-STATE CIRCUITS LETTERS

Sumikawa, Rei; Kosuge, Atsutake; Hsu, Yao-Chung; Shiba, Kota; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 1

2024

Efficient FPGA Resource Utilization in Wired-Logic Processors Using Coarse and Fine Segmentation of LUTs for Non-Linear Functions

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Li, Dongzhu; Zhao, Tianqi; Kobayashi, Kenji; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 0

2024

1.2 nJ/classification 2.4 mm² asynchronous wired-logic DNN processor using synthesized nonlinear function blocks in 0.18 μm CMOS

JAPANESE JOURNAL OF APPLIED PHYSICS

Sumikawa, Rei; Shiba, Kota; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 1

2023

A Deep Metric Learning-Based Anomaly Detection System for Transparent Objects Using Polarized-Image Fusion

IEEE OPEN JOURNAL OF THE INDUSTRIAL ELECTRONICS SOCIETY

Kosuge, Atsutake; Yu, Lixing; Hamada, Mototsugu; Matsuo, Kazuki; Kuroda, Tadahiro

被引用次数: 1

2023

Crosstalk Analysis and Countermeasures of High-Bandwidth 3D-Stacked Memory Using Multi-Hop Inductive Coupling Interface

IEICE TRANSACTIONS ON ELECTRONICS

Shiba, Kota; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 0

2023

Polyomino: A 3D-SRAM-Centric Accelerator for Randomly Pruned Matrix Multiplication With Simple Reordering Algorithm and Efficient Compression Format in 180-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Shiba, Kota; Okada, Mitsuji; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 0

2023

A 7-nm FinFET 1.2-TB/s/mm² 3D-Stacked SRAM Module With 0.7-pJ/b Inductive Coupling Interface Using Over-SRAM Coil and Manchester-Encoded Synchronous Transceiver

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shiba, Kota; Okada, Mitsuji; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 8

2023

A 10.7- μ /Frame 88% Accuracy CIFAR-10 Single-Chip Neuromorphic Field-Programmable Gate Array Processor Featuring Various Nonlinear Functions of Dendrites in the Human Cerebrum

IEEE MICRO

Kosuge, Atsutake; Hsu, Yao-Chung; Sumikawa, Rei; Hamada, Mototsugu; Kuroda, Tadahiro; Ishikawa, Tomoe

被引用次数: 1

2023

A 5.2GHz RFID Chip Contactlessly Mountable on FPC at Any 90-Degree Rotation and Face Orientation

27TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2022

Miura, Reiji; Shibata, Saito; Usui, Masahiro; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 0

2022

An Anomaly Detection System for Transparent Objects Using Polarized-Image Fusion Technique

2022 IEEE SENSORS APPLICATIONS SYMPOSIUM (SAS 2022)

Yu, Lixing; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 4

2022

A Low-power RFID with 100kbps Data Rate Employing High-speed Power Clock Generator for Complementary Pass-transistor Adiabatic Logic

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Shibata, Saito; Sawabe, Yoshiki; Shiba, Kota; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 1

2022

A 7 Gb/s Micro Rotatable Transmission Line Coupler with Deep Proximity Coupling Mode and Ground Shielding Vias

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Wang, Ximing; Kosuge, Atsutake; Hayashi, Yasuhiro; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 0

2022

A 5-GHz 0.15-mm² Collision-Avoiding RFID Employing Complementary Pass-Transistor Adiabatic Logic With an Inductively Connected External Antenna in 0.18- μ m CMOS

IEEE SOLID-STATE CIRCUITS LETTERS

Shibata, Saito; Miura, Reiji; Sawabe, Yoshiki; Shiba, Kota; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 2

2022

A bonding-less 5 GHz RFID module using inductive coupling between IC and antenna

JAPANESE JOURNAL OF APPLIED PHYSICS

Miura, Reiji; Shibata, Saito; Usui, Masahiro; Shiba, Kota; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiho

被引用次数: 0

2022

mmWave-YOLO: A mmWave Imaging Radar-Based Real-Time Multiclass Object Recognition System for ADAS Applications

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Kosuge, Atsutake; Suehiro, Satoshi; Hamada, Mototsugu; Kuroda, Tadahiho

被引用次数: 36

2022

Proximity Wireless Communication Technologies: An Overview and Design Guidelines

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kosuge, Atsutake; Kuroda, Tadahiho

被引用次数: 6

2022

A 6-Gb/s Inductively-Powered Non-Contact Connector With Rotatable Transmission Line Coupler and Interface Bridge IC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiho

被引用次数: 6

2022

A 6.5 Gb/s Shared Bus Using Electromagnetic Connectors for Downsizing and Lightening Satellite Processor System

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiho

被引用次数: 1

2022

A 16 nJ/Classification FPGA-Based Wired-Logic DNN Accelerator Using Fixed-Weight Non-Linear Neural Net

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiho

被引用次数: 12

2021

A 96-MB 3D-Stacked SRAM Using Inductive Coupling With 0.4-V Transmitter, Termination Scheme and 12:1 SerDes in 40-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Shiba, Kota; Omori, Tatsuo; Ueyoshi, Kodai; Takamaeda-Yamazaki, Shinya; Motomura, Masato; Hamada, Mototsugu; Kuroda, Tadahiho

被引用次数: 14

2021

CA2 area detection from h'ppocampa microscope images using deep earning

2021 IEEE INTERNATIONAL MIDWEST SYMPOSIUM ON CIRCUITS AND SYSTEMS (MWSCAS)

Morinaga, Shohei; Ishikawa, Tomoe; Yasui, Masato; Hamada, Mototsugu; Kuroda, Tadahiho

被引用次数: 0

2021

A 5-GHz 0.15-mm² Collision Avoidable RFID Employing Complementary Pass-transistor Adiabatic Logic with an Inductively Connected External Antenna

IEEE ASIAN SOLID-STATE CIRCUITS CONFERENCE (A-SSCC 2021)

Shibata, Saito; Miura, Reiji; Sawabe, Yoshiki; Shiba, Kota; Kosuge, Atsutake; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 1

2021

2.5D integration using inductive-coupling TSV-less miniature interposer achieving 317 Gb/s/mm², 1.2 pJ/b data-transfer

JAPANESE JOURNAL OF APPLIED PHYSICS

Shiba, Kota; Cheng, Chaoran; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 0

2020

3D system-on-a-chip design with through-silicon-via-less power supply using highly doped silicon via

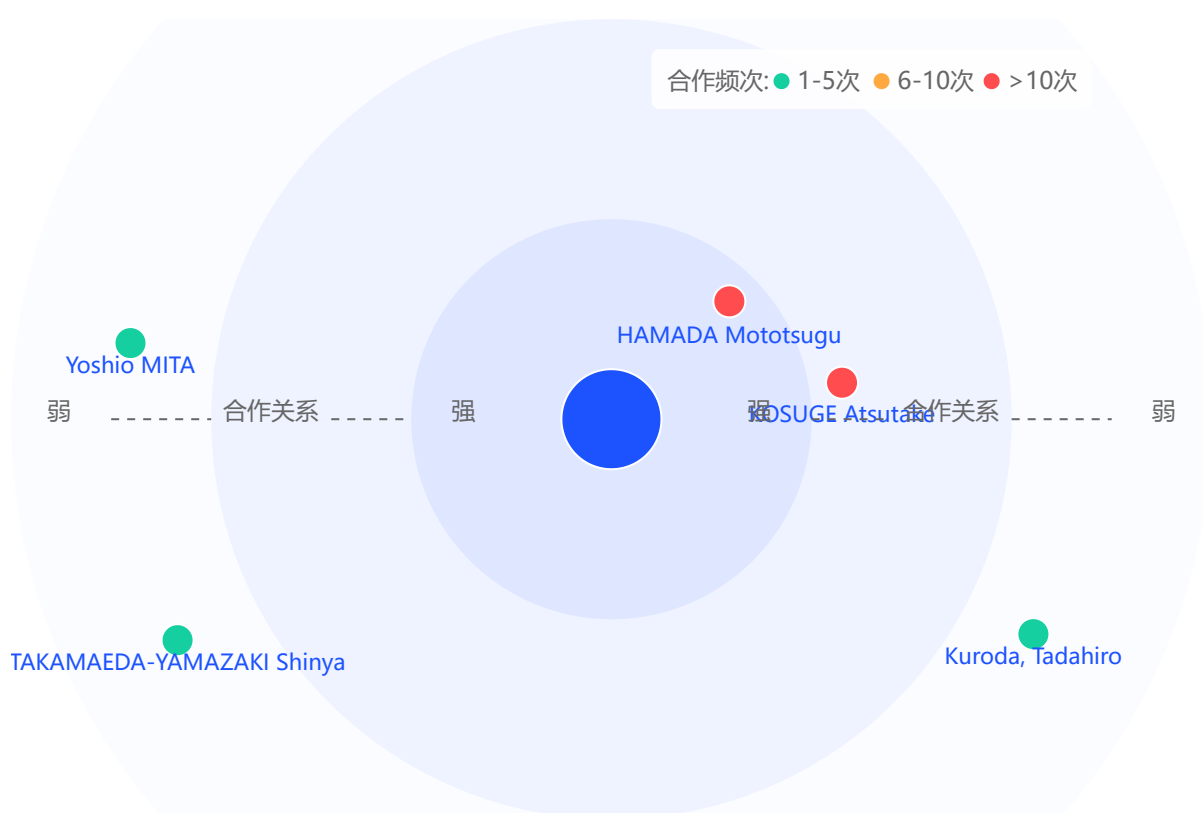
JAPANESE JOURNAL OF APPLIED PHYSICS

Shiba, Kota; Hamada, Mototsugu; Kuroda, Tadahiro

被引用次数: 3

2020

学术合作



HAMADA Mototsugu

东京大学

合作次数: 25

KOSUGE Atsutake 东京大学	合作次数: 20
Kuroda, Tadahiro 庆应义塾大学	合作次数: 2
TAKAMAEDA-YAMAZAKI Shinya 东京大学	合作次数: 1
Yoshio MITA 东京大学	合作次数: 1